

2	(a) $d = v \times t$	C1	
	$t = 0.2 \times 4$ (allow $t = 0.2 \times 2$)	C1	
	$d = 3 \times 10^8 \times 0.8 \times 10^{-6}$ OR $3 \times 10^8 \times 0.4 \times 10^{-6}$	C1	
	$d = 240$ m hence distance from source to reflector = 120 m	A1	[4]
	(b) speed of sound 300 cf speed of light 3×10^8 OR time = $240 / 300$ (= 0.8) OR time = $120 / 300$ (= 0.4)		C1
	sound slower by factor of 10^6 OR time for one division $0.8 / 4$ OR time for one division $0.4 / 2$	C1	
	time base setting 0.2 s cm^{-1} [unit required]	A1	[3]